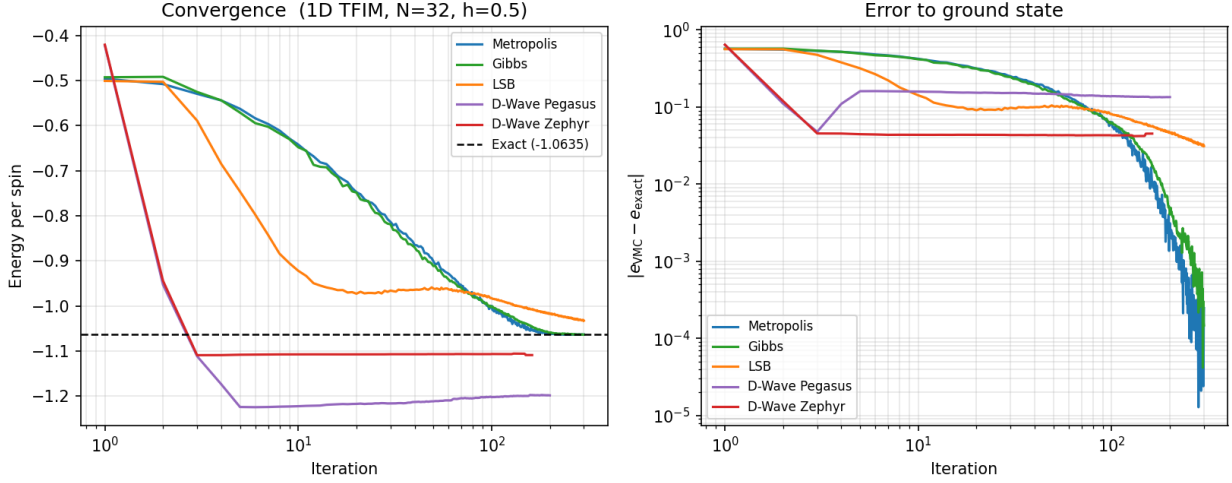


VMC/RBM Results Overview

1D TFIM and Long-Range 1D TFIM

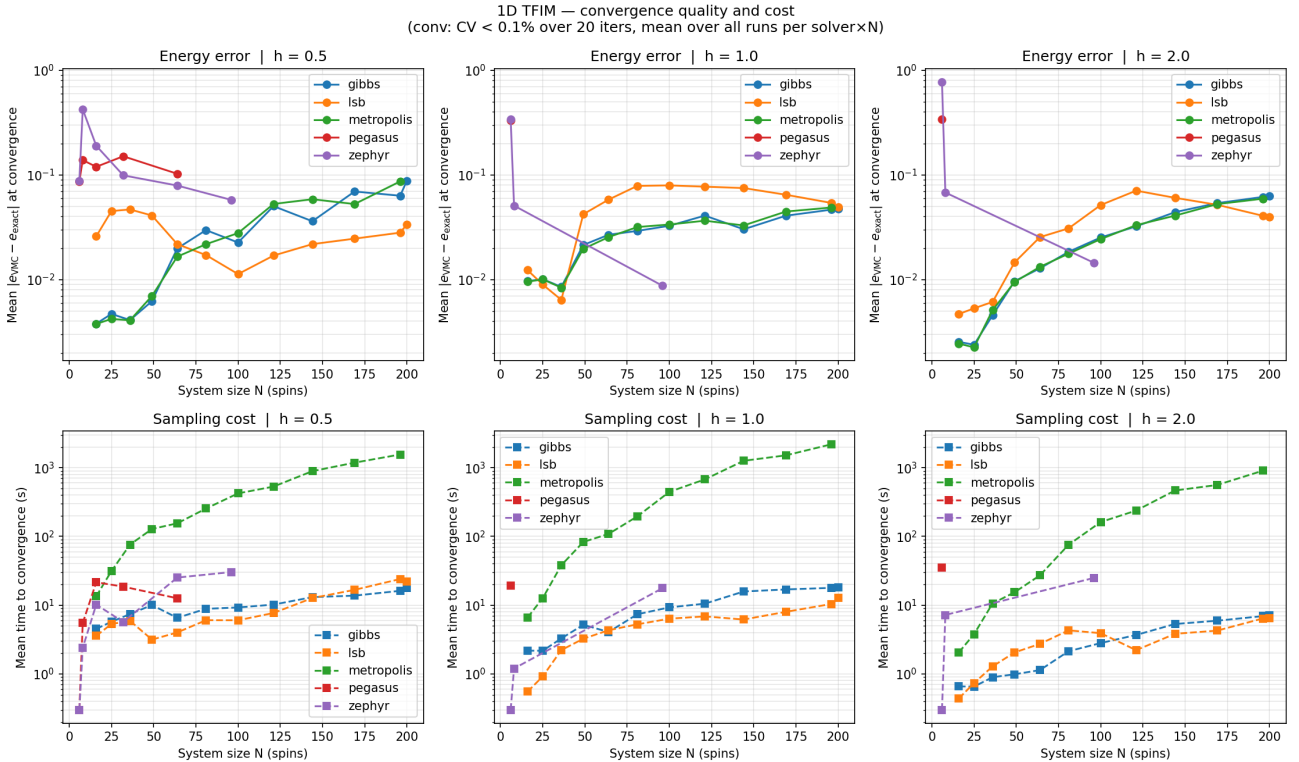
April 30, 2026

1D TFIM — Sampler Comparison (N = 32, h = 0.5, full RBM)



Energy per spin (left) and absolute error (right) for all sampler backends. Metropolis, Gibbs, and LSB converge to $\lesssim 10^{-4}$ per spin; D-Wave Pegasus and Zephyr plateau near 10^{-2} .

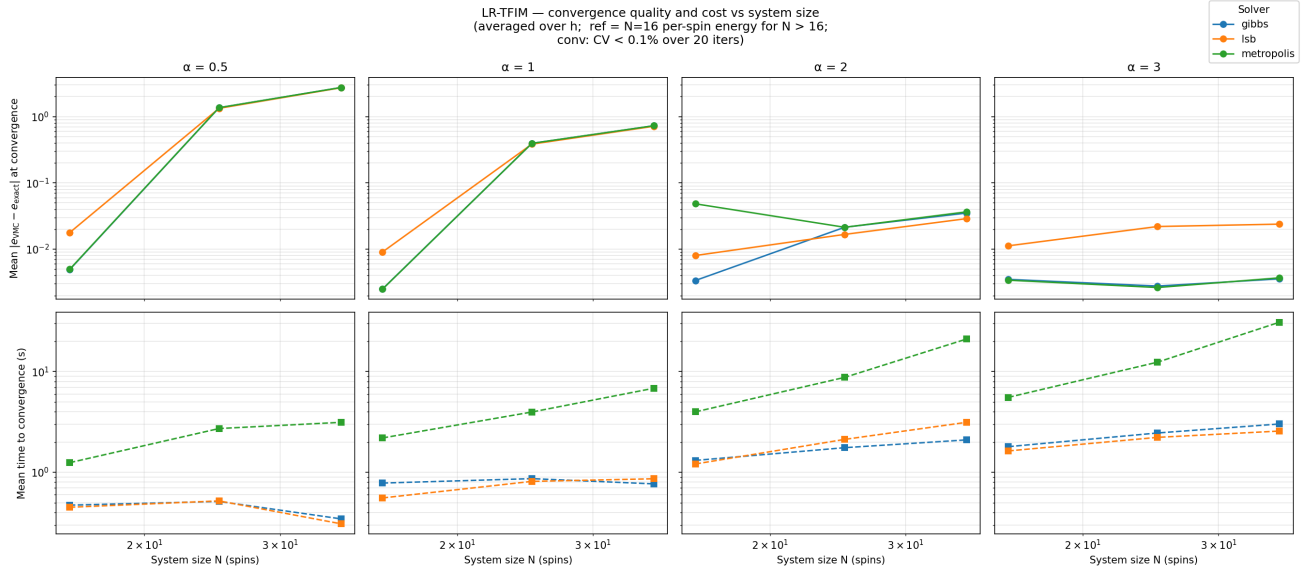
1D TFIM — Error and Cost vs System Size



Mean energy error per spin at convergence (top) and time to convergence (bottom) vs. N , for $h \in \{0.5, 1.0, 2.0\}$. Convergence: CV < 0.1% over 20 iterations.

Long-Range 1D TFIM — Error and Cost vs System Size

$H = -J \sum_{i < j} \sigma_i^z \sigma_j^z / d(i, j)^\alpha - h \sum_i \sigma_i^x$, $d(i, j) = \min(|i - j|, N - |i - j|)$. $N \in \{16, 25, 36\}$; reference energy per spin from $N = 16$ ED for $N > 16$.



LR-TFIM vs. 1D TFIM — Direct Comparison at $N = 16$

